

Pie Tactic Predictor

Fine-tuning Experiments Report

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1 Dataset

The training corpus consists of proof tactic prediction examples extracted from the Pie language. Each example provides the current proof goal together with local and global contexts, and the model must predict the correct tactic to apply.

Table 1: Dataset statistics

Statistic	Value
Total examples	7,840
Unique theorems	1,479
Unique tactic types	11
Training split	6535
Validation split	727
Test split	578

2 Models and Hyperparameters

4 candidate models spanning 1.3B to 34B parameters were evaluated. Table 2 lists each model alongside its LoRA and training hyperparameters. 0 of 4 models have completed training with valid results.

Table 2: Model catalogue and hyperparameter configuration

Model ID	HuggingFace ID	LoRA r	α	Batch	Grad. acc.	Eff. batch	LR	Epochs	Trained
deepseek-coder-1.3b	deepseek-ai/deepseek-coder-1.3b-base	64	128	16	2	32	0.0002	5	×
deepseek-coder-6.7b	deepseek-ai/deepseek-coder-6.7b-base	32	64	8	4	32	0.0001	5	×
qwen2.5-coder-1.5b	Qwen/Qwen2.5-Coder-1.5B	64	128	16	2	32	0.0002	5	×
qwen2.5-coder-7b	Qwen/Qwen2.5-Coder-7B	32	64	8	4	32	0.0001	5	×

3 Training Results

No training results available.

4 Analysis and Discussion

4.1 Completed Runs

The following models trained successfully and produced meaningful metrics: none. The best-performing model overall was N/A.

4.2 Compute Budget

Total GPU time across all completed runs was approximately **0.0 GPU-hours**.